



Fig. 2: Mercury arc lamps generate light in terahertz

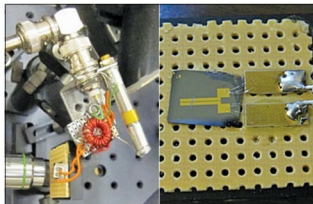


Fig. 3: Close-up of the electron source

opacity of the Earth's atmosphere to sub-millimetre radiation restricts these observatories to very high altitude sites, or to space. About a decade back the only viable sources of THz radiation were:

1. The gyrotron
2. The backward-wave oscillator (BWO)
3. The far IR laser (FIR laser)
4. Quantum cascade laser
5. The free electron laser (FEL)
6. Synchrotron light sources
7. Photo-mixing sources
8. Single-cycle sources used in THz time domain spectroscopy,

such as photo-conductive, surface field, photo-dember and optical-rectification emitters

There have also been solid-state sources of millimetre and sub-millimetre waves for many years. Nowadays, most time-domain work is done via ultra-fast lasers. In mid 2007, scientists announced the creation of a compact device that can lead to portable, battery-operated sources of

T-rays, or THz radiation. This new T-ray source uses high-temperature superconducting crystals that comprise stacks of Josephson junctions that exhibit a unique electrical property—when an external voltage is applied, an alternating current will flow back and forth across junctions at a frequency proportional to the strength of the voltage (phenomenon known as Josephson effect). These alternating currents then produce electro-magnetic fields whose frequency is tuned by the applied voltage. Even a small voltage, around two millivolts per junction, can induce frequencies in the THz range.

In 2008, engineers announced they had built a room-temperature semiconductor source of coherent THz radiation. Until then, sources

had required cryogenic cooling, greatly limiting their use in everyday applications.

Many other THz-source technologies have been investigated in the past four decades. Numerous groups worldwide are producing tuneable CW THz radiation using photo-mixing of near-IR lasers. Direct multiplied (DM) sources take millimetre-wave sources and directly multiply their output up to THz frequencies. DM sources with frequencies up to a little more than 1THz and approximately 1µW of output have been used as local oscillators for heterodyne receivers in select applications, most of which are in radio astronomy.

However, these can produce substantially more output power at lower frequencies, and are often well-suited to applications requiring frequencies of less than 500GHz. In addition, physicists have recently demonstrated quantum-cascade semiconductor lasers operating at wavelengths in the 4.4THz regime. These lasers are made from 1500 alternating layers (or stages) of gallium-arsenide and aluminium-gallium-arsenide and have produced 2mW of peak power (20nW average power), and advances in output power and operating wavelength continue at a rapid pace.

Recently, researchers at JILA (formerly known as Joint Institute for Laboratory Astrophysics), jointly operated by University of Colorado and National Institute of Standards and Technology, USA, have developed a laser based source of THz radiation that is unusually efficient and less prone to damage than similar systems. JILA instruments for generating THz radiation make use of ultra-fast pulses of near-IR laser light that enter through the lens on left, striking a semiconductor wafer studded with electrodes (transparent square that is barely visible under the white box con-

emitter arrays. These devices operate at low voltages, allowing compatibility with CMOS circuitry, coupled with innovations in antenna design and travelling wave devices. This leap in performance and flexibility enables a host of new applications for TE.

Sources of terahertz radiations

One of the main reasons that THz applications have not fully materialised yet is the lack of a small, low-cost, moderate-power THz source. THz radiations are generally emitted as a part of black-body radiation from anything with temperatures greater than 10 kelvin. While this thermal emission is very weak, observations at these frequencies are important for characterising the cold 10-20K dust in the interstellar medium. However, the